

exercise4.py

```
1 from sklearn.feature_extraction.text import CountVectorizer
2 from sklearn.metrics.pairwise import euclidean_distances
3
4 # Collect user input
5 documents = []
6 text = input("Enter document contents or type 'exit' to end input: ")
7 while text.strip() != 'exit':
8     documents.append(text.strip())
9     text = input("Enter document contents or type 'exit' to end input: ")
10 print("The euclidean distances between these documents are:")
11
12 # WRITE YOUR CODE HERE
13 vectorizer = CountVectorizer(stop_words='english', binary=True)
14 X = vectorizer.fit_transform(documents)
15 print( euclidean_distances(X) )
```

87% (13:45)

Terminal

```
codio@horizonliquid-elasticsusan:~/workspace$ python3 exercise4.py
Enter document contents or type 'exit' to end input: This is the first document.
Enter document contents or type 'exit' to end input: This document is the second document.
Enter document contents or type 'exit' to end input: exit
The euclidean distances between these documents are:
[[0. 1.]
 [1. 0.]]
codio@horizonliquid-elasticsusan:~/workspace$
```

Guide

Collapse



Coding Exercise 4

In the top-left pane, write code that will calculate and print the Euclidean Distance of the `documents` which will be captured by keyboard input.

Make sure to:

- Use `CountVectorizer`
- Filter out punctuation
- Filter out stop words
- Ignore case
- Score by existence (i.e. `binary` scoring)

Output the Euclidean Distance.

Test your code using the terminal in the bottom-left:

[TRY IT](#)

Example Input/Output:

```
This is the first document.
This document is the second document.
exit
```

```
The euclidean distances between these documents are:
[[0. 1.]
 [1. 0.]]
```

Submit your code for feedback by pressing the button below.

The first step is to properly vectorize the documents:

```
vectorizer = CountVectorizer(stop_words='english')
X = vectorizer.fit_transform(documents)
```

The second step is to calculate the distances between the documents and print them out:

```
print( euclidean_distances(X) )
```

[Check It!](#)

LAST RUN on 12/27/2023,
4:47:33 AM

Check 1 **passed**

Check 2 **passed**

Check 3 **passed**

[Next](#)